

Pseudocode

Learning Objectives

- Write clear pseudocode to represent algorithms
- Translate between pseudocode, flowcharts, and natural language
- Use structured constructs (sequence, selection, iteration) in pseudocode

Engage (5-7 minutes)

Display a complex flowchart and ask students to write instructions for it in plain English. Then show the same logic in pseudocode. Discuss how pseudocode balances readability with precision—more structured than natural language but less syntax-heavy than real code.

Explore (10-12 minutes)

Students receive a flowchart of a grade calculator (input marks, determine letter grade, output result). Working in groups, they convert it to pseudocode using keywords like INPUT, OUTPUT, IF-THEN-ELSE, WHILE, FOR, and END. Groups then swap and attempt to execute each other's pseudocode manually as a "human computer."

Explain (10-12 minutes)

Pseudocode is an informal, language-agnostic notation for expressing algorithms using structured English-like syntax. It employs keywords for control flow: sequence (consecutive statements), selection (IF/ELSE, CASE/WHEN), and iteration (WHILE, FOR, REPEAT-UNTIL). Unlike flowcharts, pseudocode is compact and easily translated into any programming language. It bridges the gap between human reasoning and machine implementation.

Elaborate (8-10 minutes)

Examine pseudocode from real-world specifications: ISO standards, API documentation, and patent filings. Compare pseudocode conventions across contexts (academic vs. industry). Discuss nested structures—how pseudocode handles multi-level decision trees and nested loops. Practice converting a mathematical formula (e.g., compound interest) into pseudocode with input validation.

Evaluate (5-8 minutes)

Students write pseudocode for a parking lot system: checking availability, assigning spots, handling payment, and issuing tickets. Assessment criteria: correct use of keywords, logical completeness,

readability, and proper termination. Self-evaluation: students compare their pseudocode to a reference solution and identify areas for improvement.